

RISK-LIMITED MEMORY REUSE: SHARP COLD-START LIMITS AND JOINT-HISTORY AUDIT SCHEDULES

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ABSTRACT

An agent can remember an observation correctly and nevertheless act incorrectly because its source has changed. We study the verification cost of controlling the probability of any stale-memory use in a finite workflow, with unknown source-change rates and truthful audits. For a source used at every step, we characterize the optimal expected audit saving on a stable environment among all uniformly reliable randomized adaptive policies. Every policy is dominated, at the same stable-instance saving, by a mixture of two adjacent policies that audit first and reuse a final block only after an unchanged history. A closed-form relaxation implies that a 5% failure budget permits at most 12.72% skipped checks without prior history. For multiple independent sources, we derive a sharp joint-history certificate governed by the maximum exposure-to-history ratio rather than the number of sources. Independent pilot observations can select source groups and audit budgets without invalidating this certificate. An exact dynamic program optimizes a restricted, discretized grouping objective. Synthetic validation includes 1,856 adaptive policies, 71,808 explicit change worlds, and 54,000 workflow rollouts. Grouping improves deployment audit savings in some heterogeneous regimes but can deteriorate with longer histories. The guarantees are marginal over history and deployment, not conditional safety for every admitted memory, and do not establish reliability of language-model agents.

1 INTRODUCTION

Persistent memory is valuable precisely because an agent can avoid reacquiring information. That saving creates a verification problem: a cached endpoint, configuration value, or execution constraint can faithfully reproduce a past observation while no longer describing the environment. Recent agent experiments identify stale inherited constraints as a cause of consequential decisions and study targeted verification interventions (Nakayashiki, 2026). Freshness-aware semantic caching also explicitly considers risk and refresh decisions (Mansoor et al., 2026). The present work asks a narrower question: *how many checks can unknown-rate memory reuse avoid while controlling the probability of any stale use during a workflow?*

This objective differs from maximizing average freshness. A workflow can have high average per-action accuracy but a substantial probability of at least one failure. It also differs from estimating that an admitted memory is safe with high confidence. A marginal guarantee averages over the history that led to admission as well as future source changes. Those two sources of randomness cannot be silently interchanged.

We isolate the problem in an auditable model: source changes are independent Bernoulli events with unknown constant rates; changed values never revert; checks are authoritative; and the workflow’s information dependencies are known. The abstraction excludes reasoning failures and natural-language retrieval. Its purpose is to establish limits on the verification controller, not to simulate an entire language-model agent.

The first result is an exact characterization, rather than only an impossibility bound. Even when the realized source is perfectly stable, a policy required to be reliable for every unknown change rate cannot immediately stop checking. We show that all randomized adaptive policies admit a

stable-instance normal form consisting of two adjacent *audit-prefix*, *reuse-tail* policies. This resolves the best possible stable-instance saving and quantifies the value of prior unchanged observations.

The second result concerns multiple sources. Whole-workflow failure depends on the union of change opportunities exposed by the audit calendar, not the sum of memory ages at every use. A gate that accepts a group only when all its historical observations are unchanged has a sharp, dimension-free marginal certificate. Grouping nevertheless has a cost: a single changing source can cause a large group to reject all reuse. We optimize this tradeoff with a pilot-informed ordered-partition dynamic program whose safety does not require accurate pilot estimates. We also give explicit failures of post-selecting a clean source, assuming independence incorrectly, and transferring a certificate across a rate shift.

Position relative to prior work. Learning change rates from intermittent checks is not new. Upadhyay et al. (2019) study binary change observations, confidence intervals, and learning-based crawling for average freshness; Avrachenkov et al. (2020) analyze change-rate estimators and freshness allocation. Kriouile and Assaad (2022) study age of incorrect information with an unknown Markovian source. Failure-free reliability demonstration likewise has a long history, including predictive and Bayesian formulations (Coolen and Coolen-Schrijner, 2005; Wilson and Farrow, 2019). Our contribution is not a new estimator, the exponential no-change probability, or the general idea of risk-gated caching. It is the exact adaptive stable-instance normal form, its connection to a sharp multi-source history gate, and selection-safe grouping under a finite-workflow failure objective. We use established fixed-sample binomial confidence bounds as a stronger-guarantee baseline (Clopper and Pearson, 1934); we do not claim optional-stopping validity, which requires different inference tools (Howard et al., 2021).

2 MODEL AND TWO RELIABILITY OBJECTIVES

Source $i \in \{1, \dots, N\}$ has unknown rate $h_i \in [0, 1]$. Independently across sources and time, its value changes with probability h_i before the action at that time. Each change produces a new, nonrepeating value. Let $\lambda_i = -\log(1 - h_i)$, with the usual endpoint convention. An audit observes the current value, updates the cache, and costs $c_i > 0$. No change occurs between an audit and the immediate action. All sources are audited at time zero; this common initialization cost is excluded from deployment comparisons.

The workflow requires source i at known times $0 < u_{i1} < \dots < u_{ij_i} \leq T$. Audits may precede uses. Every required action is executed, and a failure occurs if any required unaudited value is stale. Denote this event by F_T . Apart from audit outcomes, neither version labels nor action feedback convey extra information about h before a failure. The model does not permit bypassing the objective by abstention.

Historical data span m_i independent source-change opportunities before time zero. An unchanged pair of endpoint observations certifies that no change occurred over the interval because values do not revert. A history can contain additional intermediate audits, but a clean history means all its observations are unchanged. Therefore

$$\mathbb{P}_h(\text{source } i \text{ history is clean}) = (1 - h_i)^{m_i} = e^{-m_i \lambda_i}. \quad (1)$$

The m_i are *elapsed opportunities*, not necessarily audit counts. Audit cost, historical span, and independent pilot cost are reported separately.

Marginal workflow reliability. The central requirement is

$$\sup_{h \in [0, 1]^N} \mathbb{P}_h(F_T) \leq \delta. \quad (2)$$

The probability includes historical observations, internal randomization, and future changes. For the single-source theorem, the objective is to maximize the expected skipped checks $\mathbb{E}_0 K$ when $h = 0$. This is an instance-specific objective under a uniform risk constraint, not a minimax expected-cost claim for every environment.

High-confidence conditional safety. For a history-dependent fixed calendar $A(H)$, a different requirement is

$$\mathbb{P}_h\{H : \mathcal{R}_h(A(H)) > \rho\} \leq \alpha, \quad \mathcal{R}_h(A) = \mathbb{P}_h(F_T \mid A, H), \quad (3)$$

where future increments are independent of H . It implies a marginal bound $\alpha + (1 - \alpha)\rho$. The converse fails. Our main gates satisfy Equation (2); a successful gate is *not* a statement that its conditional failure risk is at most δ .

3 AN EXACT COLD-START NORMAL FORM

Consider one source required at every step $1, \dots, T$, with historical span m . Fix $k \in \{0, \dots, T\}$. The *tail gate* G_k audits every action in the first $T - k$ steps. If the history or any prefix audit contains a change, it audits all remaining actions. Otherwise it reuses memory for the final k actions. Write $q = 1 - h$ and $n = m + T$. Its failure probability is

$$f_k(h) = q^{n-k}(1 - q^k), \quad f_0(h) = 0. \quad (4)$$

At $h = 0$, it saves exactly k checks. This policy uses deployment audits as evidence, rather than treating the entire schedule as fixed in advance.

Theorem 1 (Exact randomized stable-instance frontier). *For any randomized adaptive policy π , let $x = \mathbb{E}_0 K$, $k = \lfloor x \rfloor$, and $a = x - k$. There is a policy $\tilde{\pi}$ that chooses G_k with probability $1 - a$ and G_{k+1} with probability a , before seeing data, such that*

$$\mathbb{E}_0 K_{\tilde{\pi}} = x, \quad \mathbb{P}_h(F_T; \tilde{\pi}) = (1 - a)f_k(h) + af_{k+1}(h) \leq \mathbb{P}_h(F_T; \pi) \quad \text{for every } h. \quad (5)$$

At $x = T$ only G_T is used. Consequently the optimal stable saving under Equation (2) is exactly

$$x^*(m, T, \delta) = \max \left\{ x \in [0, T] : \sup_h [(1 - a)f_k(h) + af_{k+1}(h)] \leq \delta \right\}. \quad (6)$$

Why an adaptive policy reduces to two tails. Fix its internal random seed and follow the trajectory with no changes. If it skips a set S of checks, a *first* change at any $t \in S$ produces an immediate failure. These events are disjoint, giving risk at least $\sum_{t \in S} hq^{m+t-1}$. With $|S| = k$, this sum is minimized by the last k times, and equals $f_k(h)$. A tail gate attains the bound. Finally $f_k(h) = q^n(q^{-k} - 1)$ is convex in k : among distributions over integer skip counts with a fixed mean, adjacent integers minimize its expectation simultaneously for every h . Appendix A gives endpoint and randomization details.

Define the increasing function

$$\psi(s) = s(1 - s)^{(1-s)/s}, \quad 0 < s < 1, \quad \psi(0) = 0, \quad \psi(1) = 1, \quad (7)$$

and let $s_\delta = \psi^{-1}(\delta)$. Direct maximization gives $\sup_h f_k(h) = \psi(k/n)$.

Corollary 2 (Closed bound and deterministic optimum). *Every uniformly reliable policy satisfies*

$$\mathbb{E}_0 K \leq \min\{T, (m + T)s_\delta\}. \quad (8)$$

The deterministic optimum is $\lfloor \min\{T, (m + T)s_\delta\} \rfloor$. The exact randomized optimum lies between this integer and the continuous bound, so the gap is less than one check. At $\delta = .05$, $s_\delta = 0.1271738264$. Thus without historical data, at least 87.28% of checks are required in expectation even on a stable source.

The frontier is computable by two scalar root searches, not by enumerating belief-state policies. The least-favorable rate is positive even though the objective is evaluated at zero rate. With $m/T \geq 1/s_\delta - 1$, every deployment check can be skipped on the stable instance, but only after the history gate is passed. Figure 1 displays this transition. The result says nothing about optimality of tail gates for expected audit cost at an arbitrary nonzero h .

4 SHARP JOINT-HISTORY CERTIFICATES

For multiple sources we restrict the reuse calendar to be fixed before its gate history is observed. A gate accepts a group only if every source in that group has a clean history; otherwise it audits every required group use. This restriction enables exact risk analysis, but is not claimed optimal among all multi-source adaptive policies.

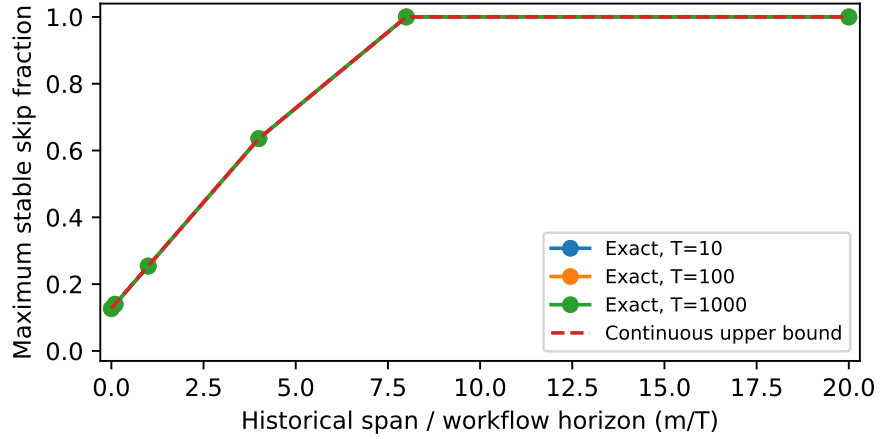


Figure 1: Exact optimal stable-instance saving at a 5% workflow failure budget. The continuous bound differs from the exact frontier by less than one check. Historical span and deployment horizon are distinct resources.

4.1 COUNT EXPOSED CHANGE OPPORTUNITIES, NOT MEMORY AGES

Let $u_{i0} = 0$, $d_{ij} = u_{ij} - u_{i,j-1}$, and $s_{ij} = 1$ when use j skips a check. Audits only at required uses suffice for fixed-calendar cost minimization: an off-use audit can be deferred to the next use without increasing cost or stale-use risk, under our unconstrained, time-independent audit costs.

Proposition 3 (Exposure identity). *For a fixed calendar,*

$$E_i = \sum_{j=1}^{J_i} d_{ij} s_{ij}, \quad \mathcal{R}_h(A) = 1 - e^{-\sum_i E_i \lambda_i}. \quad (9)$$

Here E_i is the size of the union of intervals between the last audit and each unaudited use. Repeated uses do not create independent failure events.

For example, reusing an initially current value at steps 1 through 10 exposes 10 change opportunities, not $1 + \dots + 10 = 55$. Known-rate audit minimization is a knapsack problem: skipping use (i, j) saves c_i and spends $\lambda_i d_{ij}$ from the log-risk budget $-\log(1 - \delta)$.

4.2 THE SHARP MARGINAL GATE

For a group G , write $m = (m_i)_{i \in G}$ and $E = (E_i)_{i \in G}$. Define $r = \max_{i \in G} E_i / m_i$, interpreting positive exposure with zero history as $r = \infty$, and $0/0$ as zero. Let

$$\phi(r) = \frac{r}{(1+r)^{1+1/r}} = \psi\left(\frac{r}{1+r}\right), \quad \phi(0) = 0, \quad \phi(\infty) = 1. \quad (10)$$

Theorem 4 (Joint gate, exact risk and worst case). *A fixed group and fixed calendar, accepted only after a clean history, satisfy*

$$\mathbb{P}_h(F_G) = e^{-\sum_{i \in G} m_i \lambda_i} \left(1 - e^{-\sum_{i \in G} E_i \lambda_i}\right), \quad \sup_h \mathbb{P}_h(F_G) = \phi(r). \quad (11)$$

If $0 < r < \infty$, the bound is attained by making only a maximizing-ratio source j change, with $\lambda_j = \log(1+r)/E_j$.

The proof sets $x = \sum m_i \lambda_i$ and $y = \sum E_i \lambda_i \leq rx$, then maximizes $e^{-x}(1 - e^{-rx})$. The maximizing source construction proves sharpness, not only sufficiency. Inverting the bound gives a reusable exposure allowance

$$E_i \leq r_\delta m_i \quad \text{for every } i, \quad r_\delta = \frac{s_\delta}{1 - s_\delta}. \quad (12)$$

For fixed group budget and constant per-source audit cost, an exact calendar compiler sorts each source's gaps and skips as many shortest gaps as this allowance permits. No confidence interval for each rate is needed. The dimension-free guarantee concerns *risk*; the probability of accepting the group can decay rapidly with the number of changing sources.

Proposition 5 (Price of conditional certification). *For the same gate, with nonzero exposure, Equation (3) holds uniformly exactly when*

$$r \leq \frac{-\log(1 - \rho)}{\log(1/\alpha)}. \quad (13)$$

For one source this requires $m \geq E \log(1/\alpha)/[-\log(1 - \rho)]$, whereas the marginal gate requires only $m \geq E/r_\delta$.

The difference is not a free improvement over a stronger guarantee. At the least-favorable parameter for the marginal gate, the failure probability *conditional on acceptance* is $r/(1 + r)$. At $\delta = .05$ this is approximately 12.72%, despite marginal failure at most 5%. Applications requiring every admitted memory to receive a high-confidence safety statement must use the stronger formulation.

5 LEARNING GROUPS WITHOUT INVALIDATING THE GATE

One large group shares a generous exposure allowance but rejects reuse whenever any member changes. Separate sources avoid this coupling but must split the risk budget. We use independent pilot observations to learn an intermediate grouping.

Theorem 6 (Disjoint composition and independent selection). *For disjoint groups $\mathcal{G}$ fixed independently of their gate histories, let r_G be their exposure/history ratios and S_G their accepted-calendar audit savings. Then*

$$\sup_h \mathbb{P}_h(F_T) = 1 - \prod_{G \in \mathcal{G}} (1 - \phi(r_G)), \quad (14)$$

$$\mathbb{E}_h[\text{audit saving}] = \sum_{G \in \mathcal{G}} S_G e^{-\sum_{i \in G} m_i \lambda_i}. \quad (15)$$

These statements apply conditionally on any independent pilot used to select the groups and calendars. Thus arbitrary pilot estimation error changes utility but not the uniform risk bound.

An exact restricted compiler. Let $B = -\log(1 - \delta)$ and discretize it into L units. From pilot estimates $\hat{\lambda}_i$, order sources by $m_i \hat{\lambda}_i$. Allow contiguous groups in this fixed order. A group receiving z units has $\delta_z = 1 - e^{-zB/L}$; Equation (12) determines its maximum skipped-check saving $S(a, b, z)$. Its predicted saving is

$$v(a, b, z) = S(a, b, z) \exp\left(-\sum_{i=a}^{b-1} m_i \hat{\lambda}_i\right). \quad (16)$$

The recursion

$$V(b, z) = \max_{0 \leq a < b, 0 \leq w \leq z} \{V(a, z - w) + v(a, b, w)\}, \quad V(0, z) = 0 \quad (17)$$

finds the exact best predicted saving in this ordered-partition, grid-budget class. It takes $O(N^2 L^2)$ time after source menus are formed. The output is certified because the group log budgets sum to at most B . Neither global optimality over all partitions nor optimal expected cost at the true unknown rates is asserted.

Why the independent pilot matters. If a fixed source has clean-history probability a and subsequent failure probability b , its gate risk is ab . Selecting the first clean source among N identical independent histories instead gives $[1 - (1 - a)^N]b$. With $m = 1000$, $E = 145$, and $N = 8$, a fixed-source certificate below .05 becomes actual failure .1243. The compiler must not use those same gate outcomes to decide which group to trust. Independence, not predictive accuracy, is the essential safeguard. More generally, post-selection requires a certificate for the selection procedure itself.

Method	Stable	Mixed	Active
Single group	83.3	1.9	2.0
Separate sources	41.5	31.3	26.3
Pilot-learned groups	81.7	59.1	30.4
Rate-informed groups	83.3	59.8	33.6
Joint conditional certificate	15.5	0.4	0.4
CP rate bounds	2.1	2.1	2.0
Known-rate calendar	100.0	82.4	49.0

Table 1: Mean *deployment* audit cost saved (%) at $m = 1000$, $\delta = .05$, ten seeds. Pilot and historical acquisition costs are excluded from this table and reported separately. The conditional-certificate methods provide a stronger guarantee; their smaller savings are not evidence that they are incorrectly designed.

6 SYNTHETIC EXPERIMENTS

All data are generated from specified stochastic or finite symbolic environments. There is no language-model inference, training, external dataset, or API use. The code provides exact calculations alongside an independent world-state simulator. Full parameters, numerical tolerances, and additional results appear in Appendix E.

Checks against finite oracles. We enumerated every deterministic adaptive audit policy at horizons 1 through 4: 1,856 policies in total. Polynomial maximization over $h \in [0, 1]$ verified the lower bound and an attaining tail gate for every skip count. Ten randomized-policy linear programs agreed with the exact frontier within 8×10^{-13} . These finite checks corroborate, rather than prove, Theorem 1. We also checked the exposure formula against 71,808 explicit two-source change worlds, generated 300 joint-gate and 300 composition sharpness witnesses, and tested the ordered-partition dynamic program against exhaustive small partitions. The unit suite contains 83 passing tests.

Workflow design. Each workflow has eight sources, 200 time steps, and 32 uses per source, combining clustered and scattered uses. Source-specific audit costs are one or two units; checking every required use costs 384. We vary historical span $m \in \{200, 1000, 5000\}$ and three hazard regimes: stable, mixed, and active. Ten seeds per setting perturb hazards and use times. The pilot has 64 independent windows of length 50 per source, and the compiler uses $L = 16$ budget units. We evaluate 900 method–setting decisions, integrating gate histories and future changes exactly for the clean-gate methods, and simulate 54,000 full workflows over 27 matched settings as a separate implementation check.

Comparators and probability accounting. We compare a single joint group, equal-risk singleton groups, pilot-learned groups, a rate-informed grouping reference, a joint conditional certificate with $\alpha = \rho = .025$, Clopper–Pearson (CP) rate-bound calendars, and a known-rate knapsack calendar. Point-estimated calendars and a global expiration-time baseline are also included in the raw results but are not uniformly certified. The rate-informed methods are information references, not implementable unknown-rate policies. CP calendars use ten fixed-gap historical observations and a total confidence budget .025; their CSV risks are conditional on those realized observations, unlike analytically marginalized clean-gate risks. A separate 1,200-dataset calibration experiment checks that distinction.

Gains are regime-dependent. Table 1 shows the intended tradeoff. In the mixed regime at $m = 1000$, pilot-learned groups save 59.1% of deployment audit cost, versus 31.3% for separate sources and 1.9% for a single group. Their mean exact marginal failure probability is .01784, below the uniform .05 bound. In the stable regime, one group saves 83.3%, slightly more than pilot grouping’s 81.7%. Thus learning a partition is not uniformly beneficial.

Negative results and evidence costs. Longer histories increase the permissible exposure but also make a clean gate less likely to pass. At $m = 5000$ in the stable regime, single-group savings fall to 55.4%, while separate sources save 77.8%. In the mixed regime, pilot grouping saves 44.0%,

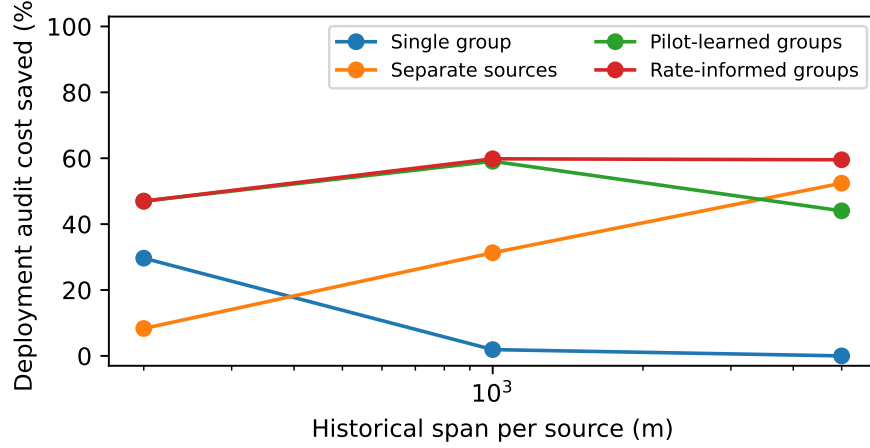


Figure 2: Mixed-rate workflows: grouping improves reuse for intermediate history spans but a noisy pilot can underperform separate-source gates at long spans. Rate-informed grouping isolates the effect of estimation error. Values integrate gate histories exactly, then average ten workflow/pilot seeds.

less than the singleton baseline’s 52.4% (Figure 2). The pilot costs 768 weighted audits, twice one fully audited workflow’s deployment cost; obtaining it solely for one workflow is not cost-effective in these experiments. A clean-history endpoint observation adds 12 units beyond the common time-zero initialization; the CP history adds 120. Our results demonstrate deployment savings given evidence, not unconditional end-to-end cost savings. Amortizing evidence across workflows does not automatically preserve a single global failure budget.

Stress tests. All clean-gate decisions obey their analytic bounds in the specified model. This does not survive arbitrary misspecification. At eight sources, post-selecting a clean source raises a nominal .05 risk to .1243; shared source changes with disjoint future exposure intervals raise it to .2765. Doubling the deployment hazard in a single-source witness gives .0933. These are constructive counterexamples, not estimated robustness guarantees.

7 LIMITATIONS AND CONCLUSION

The sharp adaptive result optimizes savings on the stable instance; the multi-source result covers fixed calendars accepted by clean-history gates. Neither solves general adaptive multi-source control. Source independence, stationary hazards, nonreverting versions, known use dependencies, and immediate truthful audits are essential. Reverting values invalidate endpoint cleanliness; correlated or shifted changes invalidate the displayed certificate unless reanalyzed. The finite experiments do not test language-model cognition or prove low failure rates in deployed agents.

Within this scope, the normal form characterizes when reliable memory reuse is possible and exactly how much stable-instance checking it can avoid. Joint histories enable a sharp multi-source gate, while independent pilot selection controls the tradeoff between shared risk budgets and group rejection. The central practical distinction is between reusing a memory under a marginal workflow guarantee and certifying that a particular admitted memory is conditionally safe.

AI USE STATEMENT

Generative AI was used extensively for ideation, literature retrieval, theorem and proof development, implementation, test design, experiment execution, analysis, and drafting. Experiments were executed by the supplied numerical simulators; no language-model calls generated experimental data or acted as experimental subjects. Numerical checks are not independent proof verification. This draft has not received independent human proof or novelty review. Before submission, the authors must review the arguments, citations, artifacts, and this disclosure.

REPRODUCIBILITY STATEMENT

The supplement contains full proofs, source code, deterministic seeds, 12 scientific CSV files with hashes, and scripts for regeneration. Main tables and figures are generated from those CSVs. Tiny instances are checked by exhaustive policy or world enumeration independently of the closed-form risk routines. The repository also records excluded acquisition costs and assumption-failure examples. No paid service or specialized accelerator is required to reproduce the experiments.

ETHICS STATEMENT

This work uses no personal data, human participants, or external systems controlled by an agent. Its guarantees should not be deployed as a substitute for source checks in high-stakes applications without validating the model and the meaning of the probability guarantee. In particular, a marginal failure budget can permit higher conditional risk for a subset of accepted histories. The paper explicitly reports this limitation rather than describing admitted memories as individually safe.

REFERENCES

- Konstantin Avrachenkov, Kishor Patil, and Gagan Thoppe. Change rate estimation and optimal freshness in web page crawling. *arXiv preprint arXiv:2004.02167*, 2020. <https://arxiv.org/abs/2004.02167>.
- C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi:10.1093/biomet/26.4.404.
- F. P. A. Coolen and P. Coolen-Schrijner. Non-parametric predictive reliability demonstration for failure-free periods. *IMA Journal of Management Mathematics*, 16(1):1–11, 2005. doi:10.1093/imaman/dph025.
- Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021. doi:10.1214/20-AOS1991.
- Saad Kriouile and Mohamad Assaad. Minimizing the age of incorrect information for unknown Markovian source. *arXiv preprint arXiv:2210.09681*, 2022. <https://arxiv.org/abs/2210.09681>.
- Muhammad Mansoor, Tahir Ahmad, and Yeo-Chan Yoon. Risk-constrained freshness-aware semantic caching for open-web retrieval-augmented LLMs. *arXiv preprint arXiv:2607.04281*, 2026. <https://arxiv.org/abs/2607.04281>.
- Kazuki Nakayashiki. When stale constraints go unchecked: Budgeted verification failures in inherited agent memory. *arXiv preprint arXiv:2608.25553*, 2026. <https://arxiv.org/abs/2608.25553>.
- Utkarsh Upadhyay, Róbert Busa-Fekete, Wojciech Kotłowski, Dávid Pál, and Balázs Szörényi. Learning to crawl. *arXiv preprint arXiv:1905.12781*, 2019. <https://arxiv.org/abs/1905.12781>.
- Kevin James Wilson and Malcolm Farrow. Assurance for sample size determination in reliability demonstration testing. *arXiv preprint arXiv:1905.08659*, 2019. <https://arxiv.org/abs/1905.08659>.

A COMPLETE PROOF OF THE ADAPTIVE NORMAL FORM

We give a pathwise argument that allows arbitrary randomized adaptive auditing. A policy can use its own random seed, all previous audit observations, and any feedback that does not reveal a source change before it causes a failure or is audited. Audit outcomes disclose version equality; labels are rate-independent. The historical observation design is fixed and spans m opportunities. A clean history is the unique history of unchanged values (up to rate-independent labels). Source hazards are constant throughout history and deployment.

A.1 FIRST-CHANGE LOWER BOUND

Condition on the policy's internal random seed $R = r$. In the no-change world $h = 0$, the history and all audit observations are deterministic, so the policy skips a fixed set $S_r \subseteq \{1, \dots, T\}$; let $K_r = |S_r|$. For $t \in S_r$, consider the event B_t that there is no change in the m historical opportunities or deployment steps $1, \dots, t-1$, and that a change occurs at step t . Its probability is $h(1-h)^{m+t-1}$, and no restriction is imposed on later changes.

On B_t , the controller has the same observable history as in the no-change world until its decision at t . Since it skips the check at t , it uses a stale value and fails. The events $\{B_t : t \in S_r\}$ are disjoint because they specify different first-change times. Hence, writing $q = 1 - h$,

$$\mathbb{P}_h(F_T \mid R = r) \geq \sum_{t \in S_r} hq^{m+t-1}. \quad (18)$$

For $0 < h < 1$ these summands decrease with t , so among sets with $K_r = k$ the last k steps minimize the sum:

$$\sum_{t \in S_r} hq^{m+t-1} \geq \sum_{t=T-k+1}^T hq^{m+t-1} \quad (19)$$

$$= q^{m+T-k}(1 - q^k) = f_k(h). \quad (20)$$

The inequality extends by continuity to $h = 0, 1$, using $0^0 = 1$ when needed. Averaging over R gives

$$\mathbb{P}_h(F_T; \pi) \geq \mathbb{E}_R f_{K_R}(h). \quad (21)$$

This argument does not assume independent repeated failure events, nor that the policy stops after a failure. Only the first stale use matters.

A.2 ATTAINMENT BY A TAIL GATE

For G_k , any historical change is detected and triggers full auditing. Any change in the first $T - k$ deployment steps is detected before that step's action and also triggers full auditing. If neither event occurs, the final k uses are unaudited and fail exactly when at least one of their source-change opportunities contains a change. History, prefix, and tail increments are independent, giving precisely

$$\mathbb{P}_h(F_T; G_k) = q^{m+T-k}(1 - q^k). \quad (22)$$

The policy executes every action. At $h = 0$, all checks are unchanged and $K = k$. Choosing k before observing data according to the distribution of K_R in Equation (21) therefore weakly reduces risk at every h without changing stable-instance saving.

A.3 TWO ADJACENT TAIL GATES SUFFICE

For fixed $0 < q < 1$, extend $f_k(h) = q^n(q^{-k} - 1)$ from integer k to real $k \in [0, T]$, where $n = m + T$. This extension is convex. Its values at integer arguments therefore have nondecreasing successive slopes. Let $\bar{f}_h$ be the piecewise-linear interpolation through these integer values. It too is convex and satisfies $\bar{f}_h(j) = f_j(h)$ for every integer j .

For $x = \mathbb{E}K = k + a$ with integer $k = \lfloor x \rfloor$ and $a \in [0, 1)$, Jensen's inequality gives

$$\mathbb{E}f_K(h) = \mathbb{E}\bar{f}_h(K) \geq \bar{f}_h(x) = (1 - a)f_k(h) + af_{k+1}(h). \quad (23)$$

Crucially the same adjacent distribution attains this lower bound for *every* h ; it is not chosen after seeing the unknown hazard. Limits give the endpoints $h = 0, 1$. This proves the domination statement of Theorem 1. Since the adjacent mixture is a valid policy, feasibility is both necessary and sufficient, establishing Equation (6). The feasible set is closed and nonempty; the risk envelope is continuous and nondecreasing in x , so the maximum exists.

A.4 CLOSED FORM, MONOTONICITY, AND ROUNDING

For real $0 < x < n$, maximizing $q^{n-x}(1 - q^x)$ over $q \in [0, 1]$ yields

$$q^x = \frac{n-x}{n}, \quad \sup_q q^{n-x}(1 - q^x) = \frac{x}{n} \left(1 - \frac{x}{n}\right)^{(n-x)/x} = \psi(x/n). \quad (24)$$

The derivative of $\log \psi(s)$ is $-\log(1 - s)/s^2 > 0$, proving strict monotonicity on $(0, 1)$. The limiting values are zero and one. Applying Jensen directly to the smooth extension gives $\mathbb{P}_h(F_T) \geq q^{n-x}(1 - q^x)$, hence $\psi(x/n) \leq \delta$ and Equation (8).

For deterministic policies $x = k$ is an integer. The tail gate is feasible exactly when $\psi(k/n) \leq \delta$, so the optimum is the claimed floor. The adjacent randomized optimum is at least this integer and at most the real upper bound. If the upper bound is an integer it is itself attainable, so the rounding gap is zero; otherwise it is strictly less than one.

As $\delta \downarrow 0$, $\psi(s) \sim s/e$, and therefore $s_\delta \sim e\delta$. The familiar first-order bound $e\delta(m + T)$ is a relaxation of the exact nonlinear expression. This asymptotic does not justify replacing the exact bound by $e\delta$ at arbitrary confidence levels.

A.5 NUMERICALLY EVALUATING THE EXACT FRONTIER

For $x = k + a$ with $0 < a < 1$, the adjacent risk as a function of q is

$$g(q) = (1 - a)q^{n-k} + aq^{n-k-1} - q^n. \quad (25)$$

Its interior stationary point solves

$$a(n - k - 1) + (1 - a)(n - k)q - nq^{k+1} = 0. \quad (26)$$

For $k \geq 1$ and $n - k - 1 > 0$, the left side is concave, positive at zero and negative at one; its positive crossing is unique. For $k = 0$, the maximizer is $q = (n - 1)/n$ when $n > 1$. For $n - k - 1 = 0$, maximize $a + (1 - a)q - q^n$ directly; if $n = 1$, the maximum is a at $q = 0$. Integer x uses the closed form. The implementation evaluates these cases with bracketed roots and includes endpoints. A second bracketed root finds the largest x with risk at most δ . The numerical implementation uses tolerances; the theorem is an exact mathematical statement, not a floating-point certificate for safety-critical code.

B EXPOSURE GEOMETRY AND CALENDAR COMPILATION

B.1 PROOF OF PROPOSITION 3

For one source, a skipped use at u_j is stale if there was a change after its last audit and at or before u_j . Collect these intervals over skipped uses. Split time into elementary intervals $(u_{j-1}, u_j]$, with $u_0 = 0$. Such an interval is in the union exactly when use j is skipped: if j is audited, its preceding changes are corrected before any later use; if it is skipped, the interval is exposed immediately. Earlier exposed intervals remain part of the union even if an audit later corrects the value. Thus the union has size $\sum_j d_j s_j$.

No stale use occurs exactly when all change indicators in this union are zero. By temporal independence this has probability $(1 - h_i)^{E_i}$ for source i . Independence across sources yields $\mathbb{P}_h(F_T^c) = \prod_i (1 - h_i)^{E_i}$. In particular, a source with $E_i = 0$ contributes a factor one even when $h_i = 1$.

An audit strictly between two consecutive required uses can be moved to the latter use. There is no intervening required action, and the later audit removes at least as many changes before the next

action. If there is already an audit there, the earlier audit can simply be removed. This argument uses no resource capacity, no time-dependent cost, and no informative value for intermediate audits in a fixed-calendar policy. It does not prove that off-use audits are redundant for general learning policies.

B.2 OPTIMAL PER-SOURCE CALENDAR UNDER A CAP

For a fixed group budget δ_G , Theorem 4 requires $E_i \leq r_{\delta_G} m_i$ separately for every source. If every check of source i has the same cost c_i , maximizing its saved cost is equivalent to maximizing the number of skipped checks. Sort d_{ij} in increasing order. For any k , the sum of the k shortest gaps is no greater than the exposure of any k skipped uses. Consequently the longest feasible prefix of this ordering is optimal. Stable sorting resolves ties without randomness. The methods allow arbitrary source costs but not arbitrary within-source time-varying costs; the latter require a knapsack solver instead of sorting.

B.3 KNOWN-RATE KNAPSACK BASELINE

With known rates, the risk constraint is $\sum_{i,j} \lambda_i d_{ij} s_{ij} \leq -\log(1 - \delta)$. For integer costs, the implementation keeps the minimum log-risk required to save each possible total integer cost, then backtracks the largest feasible saving. This is exact in the finite instances, modulo floating-point arithmetic. It avoids a risk-grid relaxation. It is optimal among fixed calendars, not among adaptive policies using audit outcomes. Its true-rate information advantage is explicitly labeled in every comparison.

C PROOFS FOR JOINT GATES AND CONDITIONAL GUARANTEES

C.1 SHARP MARGINAL CERTIFICATE

Let C_G be the event that every group history is clean. On C_G^c every required use is audited, so there is no stale-use failure. On C_G , the calendar is fixed independently of future changes. Temporal and source independence give

$$\mathbb{P}_h(F_G) = \mathbb{P}_h(C_G) \mathbb{P}_h(F_G | C_G) = e^{-x}(1 - e^{-y}), \quad x = \sum_i m_i \lambda_i, \quad y = \sum_i E_i \lambda_i. \quad (27)$$

For $0 < r < \infty$, $y \leq rx$ and therefore

$$\mathbb{P}_h(F_G) \leq e^{-x}(1 - e^{-rx}). \quad (28)$$

Differentiating with respect to $x \geq 0$ gives a unique maximum at $x^* = \log(1 + r)/r$, with value

$$e^{-x^*}(1 - e^{-rx^*}) = (1 + r)^{-1/r} \frac{r}{1 + r} = \phi(r). \quad (29)$$

Choose j with $E_j/m_j = r$, set $\lambda_j = \log(1 + r)/E_j$, and set all other rates to zero. Then $x = x^*$ and $y = rx^*$, proving attainment. If every exposure is zero, the risk is zero. If a source has positive exposure and no history, set its hazard to one and all others to zero; the risk is one. This handles the endpoint cases and proves Theorem 4.

Connection to the single-source normal form. A tail gate with k skipped deployment uses has total tested span $m + T - k$ and future exposure k . Its ratio is $r = k/(m + T - k)$, so $\phi(r) = \psi(k/(m + T))$. Thus the clean-history certificate and the adaptive normal form use the same risk function, but different policy classes: one folds prefix audits into a history; the other starts from an arbitrary adaptive policy and proves a dominating form.

C.2 PROOF OF PROPOSITION 5

Let $c = -\log(1 - \rho) > 0$. A dirty history triggers the all-audit calendar with conditional risk zero. A clean history triggers the prescribed calendar, whose risk exceeds ρ exactly when $y = \sum_i E_i \lambda_i > c$. For any fixed h in this unsafe region, the probability of issuing that unsafe calendar is e^{-x} . Since

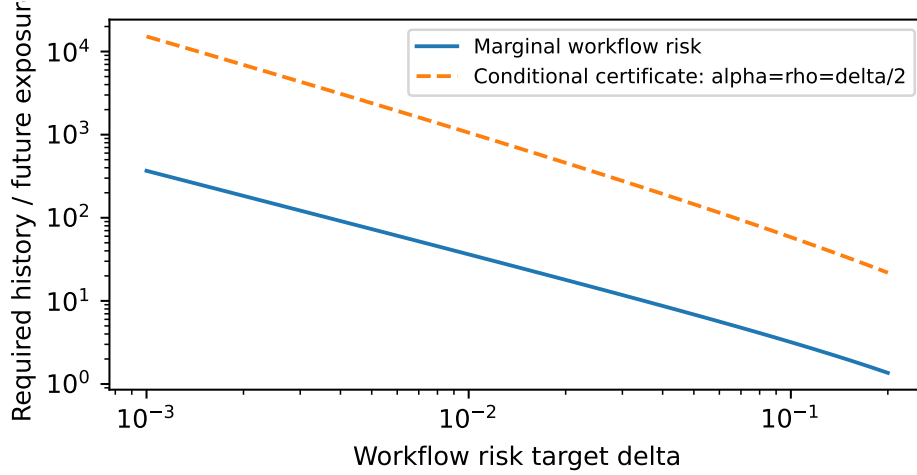


Figure 3: Historical span required per unit of future exposure. The conditional guarantee takes $\alpha = \rho = \delta/2$ and is intentionally stronger than a marginal workflow guarantee. These are evidence-span requirements, not numbers of endpoint audits.

$y \leq rx$, it is at most $e^{-c/r}$. Conversely choose a maximizing-ratio source and let $y \downarrow c$ from above. This makes $x = y/r$ and $e^{-x} \uparrow e^{-c/r}$. Hence

$$\sup_{h: \mathcal{R}_h(A) > \rho} \mathbb{P}_h(C_G) = e^{-c/r}. \quad (30)$$

The requirement that this probability be at most α is equivalent to Equation (13). The strict risk inequality does not change the supremum. The result concerns this clean-history gate, not every possible conditional-certificate procedure.

For $\alpha = \rho = \delta/2$, the conditional requirement scales as approximately $2E \log(2/\delta)/\delta$. The marginal requirement scales as $E/(e\delta)$. This separation is a comparison between different guarantees; it is not a lower bound asserting that every marginally safe algorithm needs a logarithm. Indeed Theorem 1 provides the exact stable-instance result without that logarithm.

C.3 FIXED-GAP BINOMIAL CONFIDENCE BASELINE

If audits are separated by a fixed gap d and n disjoint windows are observed, the number K of windows with at least one change is binomial with probability $p = 1 - (1 - h)^d$. The one-sided Clopper–Pearson upper endpoint is

$$p^+ = \begin{cases} \text{Beta}^{-1}(1 - \alpha; K + 1, n - K), & K < n, \\ 1, & K = n. \end{cases} \quad (31)$$

Transform it to $h^+ = 1 - (1 - p^+)^{1/d}$. In particular, $p^+ = 1$ must map to $h^+ = 1$, not a floating-point clipped smaller number. With $\sum_i \alpha_i \leq \alpha$, all true rates are covered simultaneously with probability at least $1 - \alpha$. A calendar selected using any of these same rate bounds and satisfying their worst-case future risk ρ therefore has marginal risk at most $\alpha + (1 - \alpha)\rho$.

The calendar can be data-dependent because rate coverage is simultaneous across its source coordinates. This argument does not allow arbitrary stopping times for the observations, repeated use of a fixed-sample bound, or pilot/validation leakage. We used fixed sample counts and gaps. Our separate calibration experiment reports both coverage violations and actual unsafe-calendar decisions, since the latter need not occur every time one bound misses.

D COMPOSITION, OPTIMIZATION, AND SELECTION

D.1 PROOF OF THEOREM 6

For fixed disjoint groups, F_G depends only on the histories and deployment change indicators of the sources in G . The events are therefore independent, even if sources are required by the same action. Hence $\mathbb{P}_h(F_T) = 1 - \prod_G (1 - \mathbb{P}_h(F_G))$. Each group term is bounded by $\phi(r_G)$. Conversely the separate least-favorable hazard vectors in Theorem 4 can be combined because the groups use disjoint coordinates. This attains the displayed worst case simultaneously.

The saving contributed by group G is $S_G \mathbf{1}\{C_G\}$; its expectation is $S_G e^{-\sum_{i \in G} m_i \lambda_i}$. Summing proves Equation (15). No independence is needed for linearity of the saving expectation, but independence is needed for the displayed failure product and the group acceptance formula.

Now condition on an independent pilot realization $Z = z$. The selected groups, schedules, and log budgets are deterministic functions of z (or of additional independent algorithmic randomness). The gate histories and future increments still have the original source process laws at the true fixed h . The just-proved risk inequality therefore holds conditionally on z , regardless of whether z estimates h well. Averaging over Z preserves the upper bound δ . This is an independent-data selection argument, not a claim that the pilot makes the true rates known.

D.2 OPTIMALITY OF THE RESTRICTED DYNAMIC PROGRAM

Fix a source order, grid size L , and nonnegative predicted hazards. For a contiguous group $[a, b]$ and budget z , the optimal source calendars follow the shortest-gap rule. Their predicted expected saving is $v(a, b, z)$. Any allowed partition of the first b ordered sources has a final group $[a, b]$ and spends some $w \leq z$ units on it. Its preceding groups form an allowed solution for the first a sources with at most $z - w$ units. Conversely any such preceding solution and last group form an admissible partition. Taking maxima proves Equation (17) by induction on b .

The base values $V(0, z) = 0$ permit unused budget. A zero-budget group skips no checks and contributes zero utility; thus every source can be included even when it should always be audited. There are $O(NL)$ dynamic-programming states and $O(NL)$ predecessors per state, giving $O(N^2 L^2)$ time. Group acceptance factors and skip-value sums can be calculated using prefix sums after per-source menus are formed. The implementation stores backpointers to recover the actual partition and independently recomputes its certificate before returning it.

The state space does not contain noncontiguous groups in the chosen order, nongrid budget allocations, or outcome-dependent post-gate selection. The optimality claim must not be generalized to those classes. The order and predicted utility can be poor when the pilot contains few observed changes. Table 1 and the long-history negative results illustrate the resulting performance loss.

D.3 A POST-SELECTION COUNTEREXAMPLE

Take N identical independent sources and select the first clean history, if any, reusing only that source for an exposure of E ; if none is clean, audit everything. At the fixed underlying rate h , define $a = (1 - h)^m$ and $b = 1 - (1 - h)^E$. A preselected source has risk ab . The selected source's future is independent of all histories and still has failure probability b , while the probability that some source is selected is $1 - (1 - a)^N$. Thus actual risk is

$$[1 - (1 - a)^N]b. \quad (32)$$

For $m = 1000$, $E = 145$, choose the least-favorable scalar rate $h = 1 - \exp[-\log(1 + E/m)/E]$. The fixed-source worst risk is .04977456, but Equation (32) gives .124305 at $N = 8$. Relabeling the selected source as a fixed group after the fact does not repair the guarantee.

D.4 CORRELATED SOURCES AND RATE SHIFT

For a separate counterexample, let all sources share the same Bernoulli change sequence. Give them the same historical interval of length m , so the all-clean event has probability q^m , not q^{Nm} . Construct

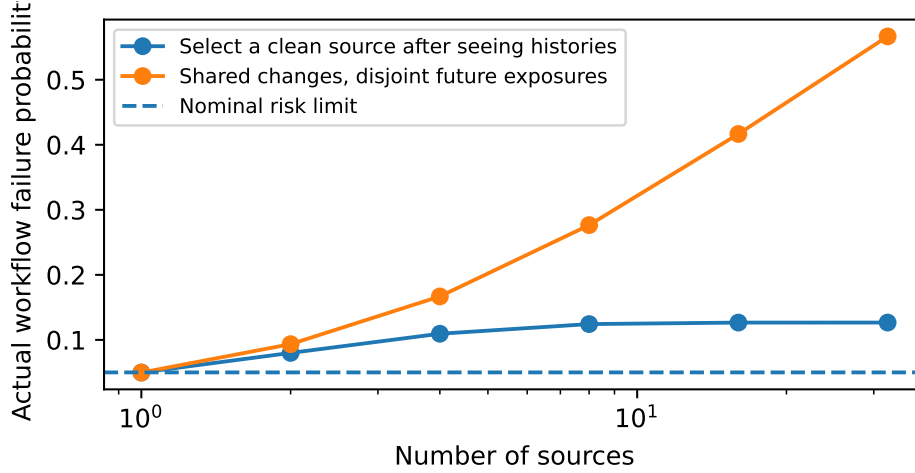


Figure 4: Constructive violations of assumptions, evaluated exactly. Selecting a source after observing the gate histories, or sharing change events across sources, can exceed the nominal 5% risk limit. These curves are not performance predictions for a deployment.

fixed calendars with pairwise disjoint future exposed intervals of length E each, with source-specific audits immediately before each interval. The actual probability of some stale use after acceptance is

$$q^m(1 - q^{NE}), \quad (33)$$

whose supremum is $\phi(NE/m)$. The independent-source formula would incorrectly give $\phi(E/m)$. This construction is deliberately outside the paper’s model; it demonstrates why source independence is not a cosmetic assumption.

A single-source time-shift example instead keeps historical hazard h_0 and increases the deployment hazard to h_1 . Actual risk is $(1 - h_0)^m[1 - (1 - h_1)^E]$. Setting h_0 to the stationary least-favorable rate and $h_1 = 2h_0$ gives .093284 for the same m, E . Our stationary certificate cannot be reused under this shift.

E EXPERIMENTAL PROTOCOL AND ADDITIONAL RESULTS

E.1 DATA GENERATION AND ACCOUNTING

Every scene has $N = 8$ sources and horizon $T = 200$. Costs are $(1, 1, 2, 2, 1, 1, 2, 2)$, so 32 checks per source cost 384. Four sources have four disjoint bursts of eight consecutive uses; burst starts are sampled from a ten-step-spaced grid. The other four have 32 distinct uniformly sampled use times. The rates before random perturbation are

$$\begin{aligned} h^{\text{stable}} &= (0, 0, 1, 1, 2, 2, 3, 3) \cdot 10^{-5}, \\ h^{\text{mixed}} &= (0, 10^{-5}, 10^{-5}, 2 \cdot 10^{-5}, 2 \cdot 10^{-5}, 3 \cdot 10^{-5}, .001, .003), \\ h^{\text{active}} &= (1, 2, 3, 4, 5, 6, 8, 10) \cdot 10^{-4}. \end{aligned}$$

Each nonzero rate is multiplied by an independent uniform factor in $[.8, 1.2]$. Ten fixed seeds generate scenes and pilots. The history span is separately set to 200, 1000, or 5000 for every source. Within a scene, methods share the same exogenous use pattern and rate vector.

The independent pilot has $n_p = 64$ windows of length $d_p = 50$ per source. If K_i windows contain changes, its plug-in hazard is $\hat{h}_i = 1 - (1 - K_i/n_p)^{1/d_p}$. These estimates only affect the grouping order and predicted utility. They are not used as confidence bounds. For the CP comparator, a second, independent history has ten windows of length $m/10$. Its confidence allocation is .025/8 per source and its future-risk budget is .025. No pilot or history parameter is tuned using hidden test changes.

For clean gates, risk and cost are analytically averaged over the second history; a history is clean precisely when none of its windows contains a change. A single observation at the beginning of

this span, followed by the common time-zero observation, is sufficient to test cleanliness under nonreversion. The extra endpoint costs 12 weighted audits; intermediate observations add no information about the *all-clean* event. CP uses the intermediate counts and is charged 120. Pilot acquisition costs 768 weighted audits and span 3200. These are fixed experimental choices, not an optimized evidence-acquisition policy. The rate-informed references receive true rates for free and are not cost-comparable deployed learners.

E.2 ORACLE AND VALIDATION COVERAGE

The 1,856 adaptive policies are all deterministic binary audit trees at horizons 1 through 4. A skip provides no change observation; an audit has clean and changed successors. We enumerate all binary source-change sequences for each policy and aggregate failure probabilities by polynomial coefficients in h . Derivative roots and interval endpoints yield an independent numerical maximizer of this polynomial. Grouping policies by stable skip count verifies both the lower envelope and attainment by a tail gate.

For randomized-policy validation, ten linear programs mix the full policy sets at horizons 3 and 4. They enforce risk bounds on a grid of 121 hazard values *and* the exact adjacent-mixture least-favorable hazard. The resulting upper relaxation matches the analytic optimum to within 8×10^{-13} . It remains a finite numerical check, not a proof of continuum feasibility. Sixty independently generated two-source tiny calendars enumerate a total of 71,808 change worlds, directly updating version counters and cache states. Their exact weighted failure frequencies match Equation (9) to numerical tolerance.

The 300 joint and 300 partition witnesses use independently sampled histories, exposures, and source counts up to six. Each checks that the explicit worst-case parameter attains the closed form. Unit tests also exhaust tiny skip subsets, knapsack solutions, and five small ordered-partition problems. All 83 tests passed locally. The suite checks zero exposure, zero history, hazard one, all-changed CP histories, and very small clean-history probabilities. Assertions use numerical tolerances; none is a machine-checked formal proof.

E.3 EXPLICIT WORKFLOW SIMULATION

For each of three regimes, three history spans, and three methods (single group, separate sources, and pilot grouping), the simulator runs 2,000 independent workflows: 54,000 total. It samples clean histories, then simulates every source change, audit update, and required use by version counters. Identical random seeds couple exogenous events across methods where possible, but controllers never access hidden states. Whole workflows, not their action steps, are independent sampling units. Figure 5 compares observed failure frequency with analytic probability. The largest absolute discrepancy is about .0059; the paper’s rare-event claims rely on the mathematical model, not on absence of observed failures.

The CP calibration uses four sources with 16 uses each, history spans 100, 1000, and 5000, two rate scales, and 200 independent histories per combination: 1,200 datasets. We separately record whether any rate bound misses and whether the selected calendar has actual conditional risk above .025. No unsafe conditional calendar was observed in this run. This observation is weaker than a formal universal claim; the valid fixed-sample guarantee follows from the confidence construction, subject to the model assumptions.

E.4 ADDITIONAL GROUPING CURVES AND WEAKER BASELINES

Figures 6 and 7 complement the mixed-rate main plot. The pilot frequently sees no changes at low rates, so many estimated hazards tie. At long historical spans, grouping errors become expensive because one genuinely active source suppresses reuse for its entire group. Rate-informed grouping has a better predicted objective because its predictions are correct, but is still restricted to an ordered partition and a log-budget grid.

The point-estimated knapsack and global expiration-time comparators spend risk according to pilot rates without a confidence correction. Their low audit costs do not imply uniform reliability. All their actual risks, including violations, are retained in `workflow_results.csv`; they are not dropped

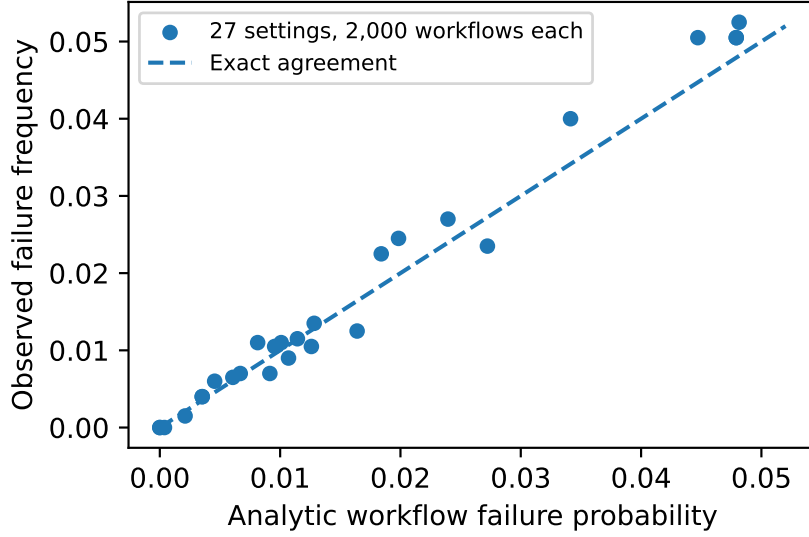


Figure 5: Independent state-simulator check of the risk calculation. Each point uses 2,000 complete workflows. This is a mechanism validation, not a claim that small samples establish extremely small failure probabilities.

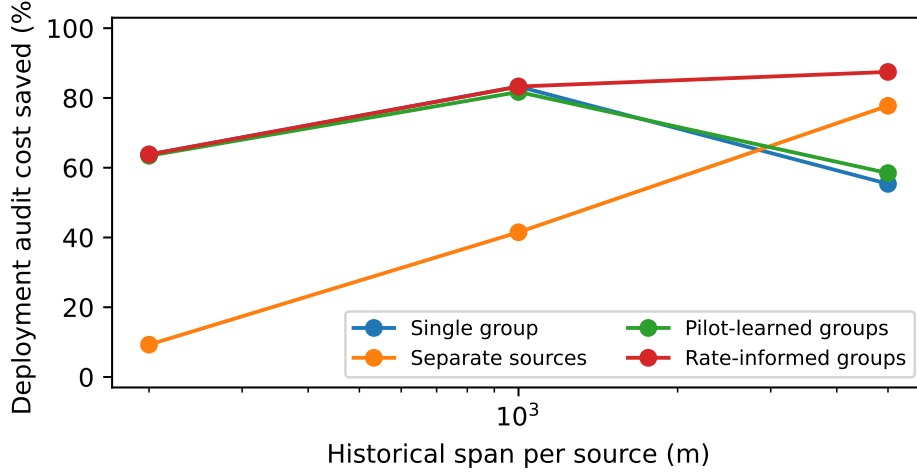


Figure 6: Stable regime. Large groups are effective with intermediate spans; separate sources can outperform a noisy learned partition with longer histories.

because they fail certification. The global expiration-time comparator is selected from all integer expiration times up to $T + 1$ using nominal risk, not hidden actual risk. Equal-source groups are a strong certified baseline that requires no learned partition or pilot acquisition.

E.5 REPRODUCTION COMMANDS AND SCOPE OF ARTIFACTS

From a clean Python environment in the repository root:

```
python -m pip install -r requirements.txt
python -m pytest -q
python experiments/run_all.py
python experiments/additional_checks.py
python experiments/make_figures.py
make paper
```

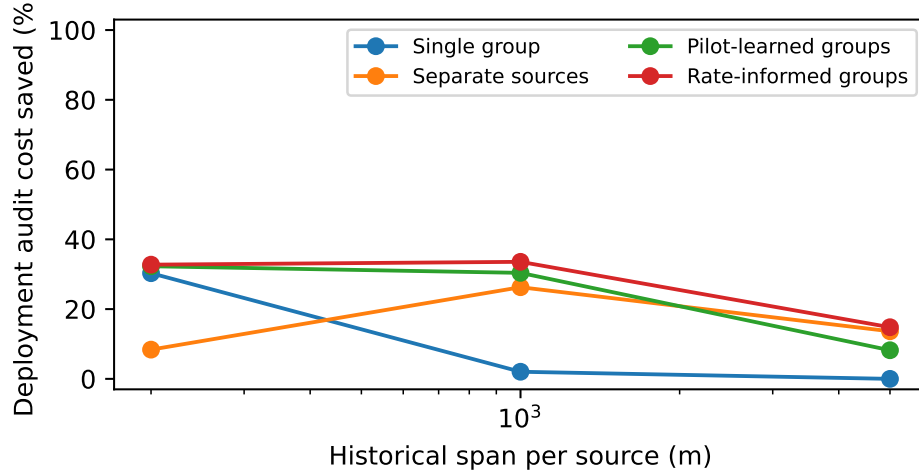


Figure 7: Active regime. A single all-clean gate has rapidly collapsing acceptance probability as the historical span grows. Even pilot grouping may lose to separate sources.

The first two experiment scripts produce 12 CSVs; the figure script produces vector PDF figures, PNG previews, the main table, and a headline JSON. CSV hashes are recorded in `results/manifest.json`. Scientific results should reproduce numerically across supported library versions; exact bytes are checked when the runtime is identical. Wall-clock metadata are not deterministic and are not scientific outcomes. The experiments require only CPU numerical libraries, not a GPU, model weights, paid APIs, or downloads. Fetching the conference style, when absent, is a separate formatting step and has no role in the experiments.

F CLAIM BOUNDARIES AND UNRESOLVED EXTENSIONS

What is established. Theorem 1 concerns every randomized adaptive policy in the single-source, every-step-use model, but optimizes only its stable-instance saving. Theorem 4 is an exact worst-case risk characterization for the stated fixed-calendar clean-history gate. Theorem 6 composes disjoint source groups; the dynamic program is exact only for its ordered-partition and budget-grid objective. The experiments check finite mechanisms and implementation consistency.

What is not established. The paper does not prove the optimal expected cost at arbitrary nonzero hazards, optimal online multi-source revalidation, or minimax-optimal pilot allocation. It does not provide a safety guarantee after conditioning on every admitted history, under arbitrary source dependence, under hazard drift, after adaptive stopping of a fixed-sample history, or for repeated workflows without risk accounting. The workflow semantics treat stale versions as action failure by design. A real task may tolerate some stale facts and fail for reasons unrelated to memory; these require different loss and observation models.

Why more data can appear harmful. For a fixed clean-history gate, more history simultaneously increases $E \leq r_\delta m$ and decreases the probability of acceptance. A policy allowed to select a shorter *prespecified* subhistory can avoid some losses, but choosing the most favorable clean subhistory after seeing all outcomes is a selection problem. Our comparisons hold the historical window fixed and reveal that gate-specific tradeoff. They do not prove that access to additional data reduces the optimal attainable performance.

Amortization and future directions. Historical evidence acquired for other purposes can make reuse worthwhile, but purchasing a pilot for one short workflow may not. Reusing a pilot with new, properly specified gate histories preserves per-workflow safety; a global multi-workflow target needs its own risk allocation or joint analysis. Optimal evidence acquisition and outcome-rich testing rules remain outside our guarantees.